COHOMOLOGICAL DESCENT FOR PROPER HYPERCOVERINGS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-compare-simplicial-objects-ph-etale, lemma-direct-image-morphism-simplicial-sites, lemma-hypercovering-X-simple-descent-bounded-abelian; external tools from More on Cohomology of Spaces and Topologies on Spaces.

Editorial note. The label lemma-cohomological-descent-for-ph-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Comparison diagram. Same unit, Lemma lemma-compare-simplicial-objects-ph-etale. The commutative diagram relating a, h, a_ph used to move between the etale and ph topologies.
- Proper hypercovering descent on components. More on Cohomology of Spaces, Lemma spaces-more-cohomology-lemma-cohomological-descent-etale-ph. Rh_{n,*}h_n^{-1} is the identity on D^+(U_{n,etale}); also used for the final collapse.
- Simplicial direct image. Same unit, Lemma lemma-direct-image-morphism-simplicial-sites. Rh_*h^{-1} is the identity on D^+(U_etale) follows levelwise.
- Descent for ph hypercoverings. Same unit, Lemma lemma-hypercovering-X-simple-descent-bounded-abelian. Third equality: Ra_{ph,*}a_{ph}^{-1} is the identity after the U is a ph hypercovering (Topologies on Spaces lemma for the hypothesis).

Source reference. Original reference 0DAF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5910–5910. Prerequisite pointer invoked at proof line(s) 8166; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DHJ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7973–7973. Prerequisite pointer invoked at proof

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line(s) 8144; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D97: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1353–1353. Prerequisite pointer invoked at proof line(s) 8151; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D8N: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5086–5086. Prerequisite pointer invoked at proof line(s) 8167; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 8140; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGU: sources/stacks-chow/tags.txt, source lines 16884–16884. Prerequisite pointer invoked at proof line(s) 8148; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DFI: sources/stacks-chow/tags.txt, source lines 16838–16838. Prerequisite pointer invoked at proof line(s) 8170; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is a proper hypercovering of X, then for $K \in D^+(X_{étale})$*

$$K \to Ra_*(a^{-1}K)$$

is an isomorphism. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. Consider the diagram of Lemma 0DHJ. Observe that $Rh_{n,*}h_n^{-1}$ is the identity functor on $D^+(U_{n,étale})$ by More on Cohomology of Spaces, Lemma 0DGU. Hence Rh_*h^{-1} is the identity functor on $D^+(U_{étale})$ by Lemma 0D97. We have

$$\begin{aligned} Ra_*(a^{-1}K) &= Ra_*Rh_*h^{-1}a^{-1}K \\ &= Rh_{-1,*}Ra_{ph,*}a_{ph}^{-1}(h_{-1})^{-1}K \\ &= Rh_{-1,*}(h_{-1})^{-1}K \\ &= K \end{aligned}$$

The first equality by the discussion above, the second equality because of the commutativity of the diagram in Lemma 0DAF, the third equality by Lemma 0D8N as U is a hypercovering of X in $(Spaces/S)_{ph}$ by Topologies on Spaces, Lemma 0DFI, and the last equality by the already used More on Cohomology of Spaces, Lemma 0DGU. □